Need for Concurrency Locking Protocols

1. Introduction

locking protocols are used — they act as a **synchronization mechanism** that controls how multiple transactions access the same data item.

Need of Locking Techniques

When multiple transactions run at the same time, they may try to read or modify the same data item.
To maintain **consistency**, **correctness**, and **isolation**, DBMS uses **locking techniques**.

What is a Locking Protocol?

A **locking protocol** is a set of rules that determines **when and how a transaction can lock or unlock** a data item.
Locks ensure that **only one transaction at a time** can modify a particular piece of data, thus maintaining **consistency** and **isolation**.

Definitions

Lock is the most Common type of concurrency control mechanism. In this approach any data is Retrieved by an active user then updating must be locked or denied to other user until updating is not complete.

T'1	T'2	Result
read_lock (Y);	read_lock (X);	T1 and T2 follow two-phase
read_item (Y);	read_item (X);	policy but they are subject to
write_lock (X);	Write_lock (Y);	deadlock, which must be
unlock (Y);	unlock (X);	dealt with.
read_item (X);	read_item (Y);	
X:=X+Y;	Y:=X+Y;	
write_item (X);	write_item (Y);	
unlock (X);	unlock (Y);	

In Locking Transaction Protocols there are basically 3 types of Transaction are used:

1. Binary Locks
2. Shared/Exclusive Locks
3. 2-Phase Locking Techniques

1. **Binary Locks**
   - A binary lock has only two states [locked (1) or unlocked (0)]
   - If an object is locked by a transaction (lock on x=1) then ,No Other Transaction can use that object .
   - If an object is unlocked (lock on x=0) an transaction can lock the object it's used .
   - A **Binary Lock** allows only two lock states on a data item:
     - **0 → Unlocked**
     - **1 → Locked**
     - A binary lock does not differentiate between read or write.
       If a data item is locked, **no other transaction** can read or write it.
2. **Shared/Exclusive Lock (S/X Lock)**

   Shared/Exclusive locking is more advanced than binary locks.

   There are **two types of locks**:

(a) Shared Locks

A **Shared Lock** allows a transaction to **read** a data item, but **not modify** it.

- Multiple transactions can **hold shared locks** on the same data item **simultaneously**, since reading does not change the data.
- No transaction can obtain an **exclusive lock** on that item until all shared locks are released.

(b) Exclusive Lock (X-Lock)

An **Exclusive Lock** allows a transaction to both **read and write** a data item.

- Only **one transaction** can hold an exclusive lock on a data item at a time.
- No other transaction can obtain either **shared** or **exclusive** lock on that same item until it is released.

Notation: X(X) → Exclusive lock on data item X.

Example of S/X Locks

Transaction T1: Read balance of Account A

Transaction T2: Update balance of Account A

T1: S-Lock(A) → Allowed

T1: READ(A)

T2: X-Lock(A) → Not allowed (A already has S-lock)

T2 waits…

T1: UNLOCK(A)

T2: X-Lock(A) → Allowed now

T2: WRITE(A)

T2: UNLOCK(A)

Problem in S/X Locking

1) May not Sufficient to produce only serializable schedule.
2) May Not Free from Irrecoverability.
3) May Not Free from Deadlock.
4) May not free from starvation.

2-Phase Locking (2PL)

Two-Phase Locking (2PL) is a locking protocol used to guarantee conflict serializability and prevent inconsistent executions.

It divides the entire locking process into two phases:

3.1 Growing Phase

- Transaction **acquires** locks
- It **cannot release** any lock
- Only lock acquisition is allowed

3.2 Shrinking Phase

- Transaction **releases** locks
- It **cannot acquire** new locks
- Only lock release is allowed

Example of Two-Phase Locking (2PL)

Transactions

- **T1:** Read(A), Write(A)
- **T2:** Read(B), Write(B)

T1: X-Lock(A) ← Growing Phase

Example of 2PL

T1: READ(A)

T1: WRITE(A)

T1: UNLOCK(A) ← Shrinking Phase

(No new locks allowed)

T2: X-Lock(B)

T2: READ(B)

T2: WRITE(B)

T2: UNLOCK(B)

Real-Life Example of Shared/Exclusive Locks (S/X Locks)

Scenario:

A library reading room with the same book available for reading at the table.

Shared Lock (S):

Multiple students can read the same book at the same time,
as long as no one writes or edits the book.

- Student A reads the book → S Lock
- Student B reads the same book → S Lock allowed
- Student C wants to read → S Lock allowed

Reading is allowed simultaneously.

Exclusive Lock (X):

If someone wants to edit/write in the book (example: librarian updating the book):

- Librarian needs Exclusive Lock
- No one else can read or write at the same time

Real-Life Example of Two-Phase Locking (2PL)

Scenario:

Updating a **student mark sheet** in a college office.

The office staff must access multiple registers (Attendance Register, Marks Register).

In 2PL, there are two phases:

A. Growing Phase (Collecting All Registers)

The teacher first collects all necessary registers:

1. Takes Attendance Register → lock acquired
2. Takes Marks Register → lock acquired

During this time:

- Teacher **cannot return** any register
- Only **takes more** if needed
- This ensures no other teacher can use them

This is the **growing phase** (acquiring locks).

B. Shrinking Phase (Returning All Registers)

After calculating and filling marks:

1. Teacher returns Attendance Register
2. Teacher returns Marks Register

After the first register is returned:

- Teacher **cannot take any new register**
- Only release operations allowed

This is the **shrinking phase** (releasing locks).